



commodore mos technology NMOS

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6526 COMPLEX INTERFACE ADAPTER (CIA)

DESCRIPTION

The 6526 Complex Interface Adapter (CIA) is a 65XX bus compatible peripheral Interface device with extremely flexible timing and I/O capabilities.

FEATURES

- 16 Individually programmable I/O lines
- 8 or 16-Bit handshaking on read or write
- 2 independent, linkable 16-Bit interval timers
- 24-hour (AM/PM) time of day clock with programmable alarm
- 8-Bit shift register for serial I/O
- 2TTL Load capability
- CMOS compatible I/O lines
- 1 or 2 MHz operation available

ORDERING INFORMATION

MXS 6526

FREQUENCY RANGE
NO SUFFIX = 1 MHz
A = 2 MHz

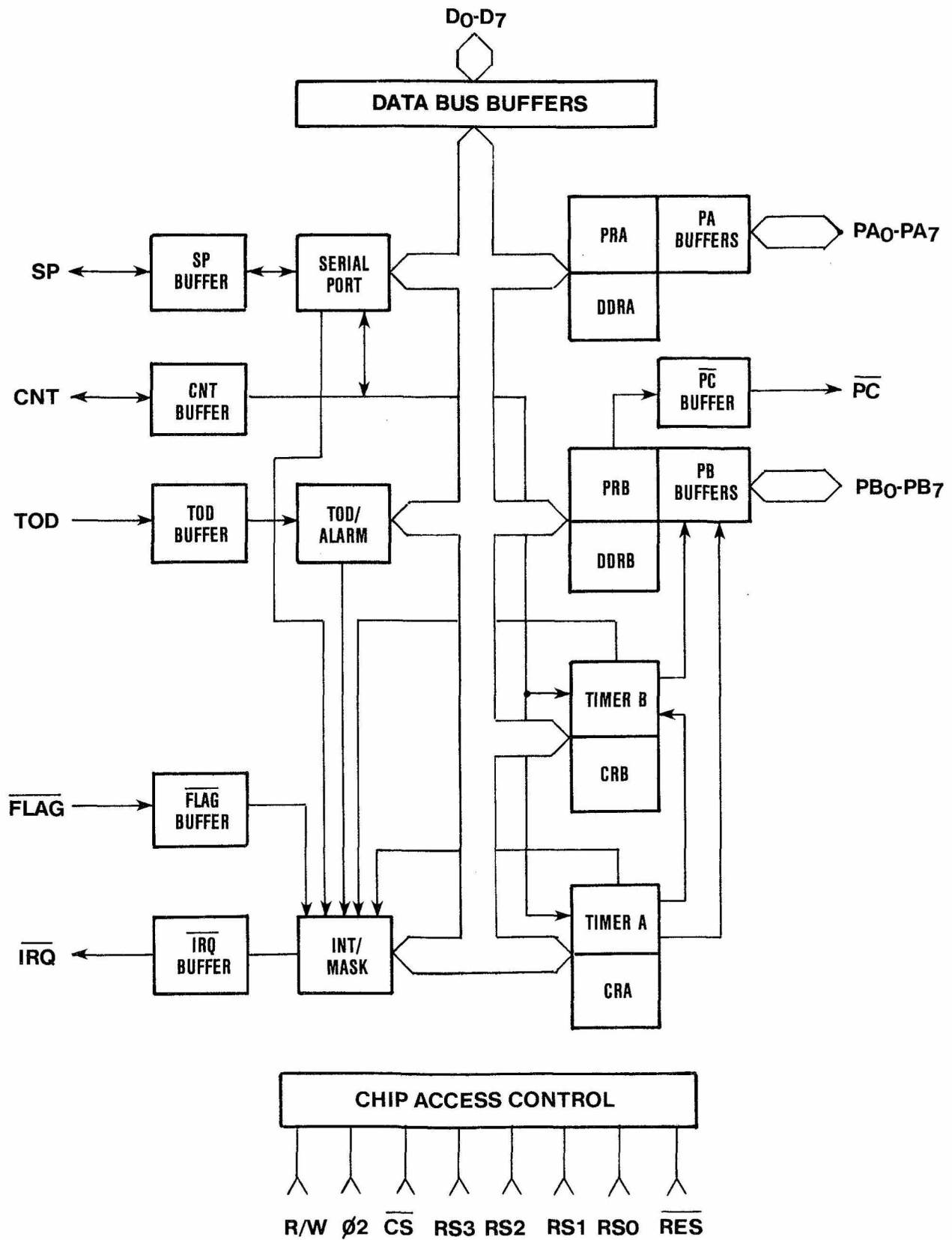
PACKAGE DESIGNATOR
C = CERAMIC
P = PLASTIC

PIN CONFIGURATION

VSS	1	40	CNT
PA0	2	39	SP
PA1	3	38	RS0
PA2	4	37	RS1
PA3	5	36	RS2
PA4	6	35	RS3
PA5	7	34	RES
PA6	8	33	DB0
PA7	9	32	DB1
PB0	10	31	DB2
PB1	11	30	DB3
PB2	12	29	DB4
PB3	13	28	DB5
PB4	14	27	DB6
PB5	15	26	DB7
PB6	16	25	Ø2
PB7	17	24	FLAG
PC	18	23	CS
TOD	19	22	R/W
VCC	20	21	IRQ



6526
BLOCK DIAGRAM



MAXIMUM RATINGS

Supply Voltage, V_{CC}	-0.3V to +7.0V
Input/Output Voltage, V_{IN}	-0.3V to +7.0V
Operating Temperature, T_{OP}	0°C to 70°C
Storage Temperature, T_{STG}	-55°C to 150°C

All inputs contain protection circuitry to prevent damage due to high static discharges. Care should be exercised to prevent unnecessary application of voltages in excess of the allowable limits.

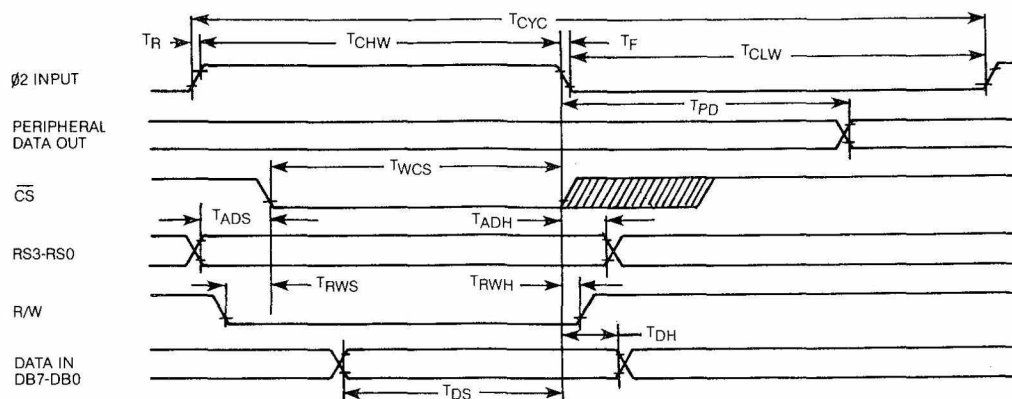
COMMENT

Stresses above those listed under "Absolute Maximum Ratings" may cause permanent damage to the device. These are stress ratings only. Functional operation of this device at these or any other conditions above those indicated in the operational sections of this specification is not implied and exposure to absolute maximum rating conditions for extended periods may affect device reliability.

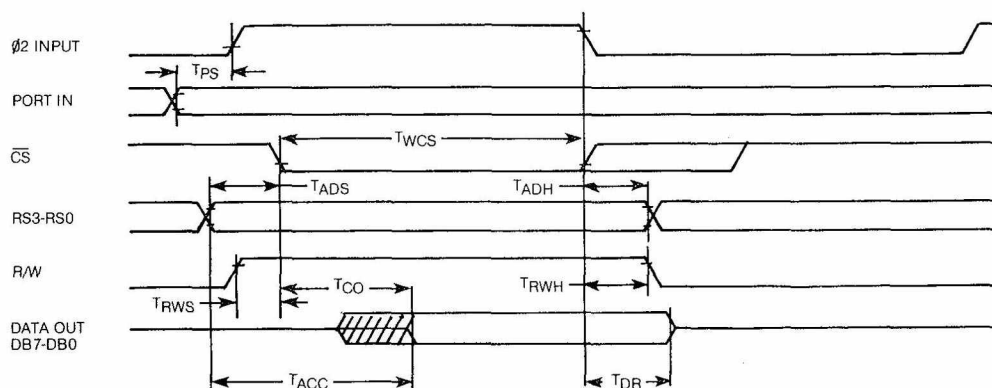
ELECTRICAL CHARACTERISTICS ($V_{CC} \pm 5\%$, $V_{SS} = 0V$, $T_A = 0-70^\circ C$)

Characteristic	Symbol	Min.	Typ.	Max.	Unit
Input High Voltage	V_{IH}	+ 2.4	—	V_{CC}	V
Input Low Voltage	V_{IL}	- 0.3	—	—	V
Input Leakage Current; $V_{IN} = V_{SS} + 5V$ (TOD, R/W, FLAG, $\overline{\phi 2}$, RES, RS0-RS3, $\overline{CS}$)	I_{IN}	—	1.0	2.5	μA
Port Input Pull-up Resistance	R_{PI}	3.1	5.0	—	$k\Omega$
Output Leakage Current for High Impedance State (Three State); $V_{IN} = 4V$ to 2.4V; (DB0-DB7, SP, CNT, IRQ)	I_{TSI}	—	± 1.0	± 10.0	μA
Output High Voltage $V_{CC} = MIN$, $I_{LOAD} < -200\mu A$ (PA0-PA7, $\overline{PC}$, PB0-PB7, DB0-DB7)	V_{OH}	+ 2.4	—	V_{CC}	V
Output Low Voltage $V_{CC} = MIN$, $I_{LOAD} < 3.2mA$	V_{OL}	—	—	+ 0.40	V
Output High Current (Sourcing); $V_{OH} > 2.4V$ (PA0-PA7, PB0-PB7, PC, DB0-DB7)	I_{OH}	-200	-1000	—	μA
Output Low Current (Sinking); $V_{OL} < .4V$ (PA0-PA7, $\overline{PC}$, PB0-PB7, DB0-DB7)	I_{OL}	3.2	—	—	mA
Input Capacitance	C_{IN}	—	7	10	pf
Output Capacitance	C_{OUT}	—	7	10	pf
Power Supply Current	I_{CC}	—	70	100	mA

6526 WRITE TIMING DIAGRAM



6526 READ TIMING DIAGRAM



6526 INTERFACE SIGNALS

$\phi 2$ — Clock Input

The $\phi 2$ clock is a TTL compatible input used for internal device operation and as a timing reference for communicating with the system data bus.

$\overline{CS}$ — Chip Select Input

The $\overline{CS}$ input controls the activity of the 6526. A low level on $\overline{CS}$ while $\phi 2$ is high causes the device to respond to signals on the R/W and address (RS) lines. A high on $\overline{CS}$ prevents these lines from controlling the 6526. The $\overline{CS}$ line is normally activated (low) at $\phi 2$ by the appropriate address combination.

R/W — Read/Write Input

The R/W signal is normally supplied by the micro-processor and controls the direction of data transfers of the 6526. A high on R/W indicates a read (data transfer out of the 6526), while a low indicates a write (data transfer into the 6526).

RS3-RS0 — Address Inputs

The address inputs select the internal registers as described by the Register Map.

DB7-BD0 — Data Bus Inputs/Outputs

The eight data bus pins transfer information between the 6526 and the system data bus. These pins are high impedance inputs unless $\overline{CS}$ is low and R/W and $\phi 2$ are high, to read the device. During this read, the data bus output buffers are enabled, driving the data from the selected register onto the system data bus.

$\overline{IRQ}$ — Interrupt Request Output

$\overline{IRQ}$ is an open drain output normally connected to the processor interrupt input. An external pullup resistor holds the signal high, allowing multiple $\overline{IRQ}$ outputs to be connected together. The $\overline{IRQ}$ output is normally off (high impedance) and is activated low as indicated in the functional description.

$\overline{RES}$ — Reset Input

A low on the $\overline{RES}$ pin resets all internal registers. The port pins are set as inputs and port registers to zero (although a read of the ports will return all highs because of passive pullups). The timer control registers are set to zero and the timer latches to all ones. All other registers are reset to zero.

6526 TIMING CHARACTERISTICS

Symbol	Characteristic	1MHz		2MHz		Unit
		MIN	MAX	MIN	MAX	
ϕ_2 Clock						
T _{CYC}	Cycle Time	1,000	20,000	500	20,000	nS
T _R , T _F	Rise and Fall Time	—	25	—	25	nS
T _{CHW}	Clock Pulse Width (High)	420	10,000	200	10,000	nS
T _{CLW}	Clock Pulse Width (Low)	420	10,000	200	10,000	nS
Write Cycle						
T _{PD}	Output Delay From ϕ_2	—	1,000	—	500	nS
T _{WCS}	CS low while ϕ_2 high	420	—	200	—	nS
T _{ADS}	Address Setup Time	0	—	0	—	nS
T _{ADH}	Address Hold Time	10	—	5	—	nS
T _{RWS}	R/W Setup Time	0	—	0	—	nS
T _{RWH}	R/W Hold Time	0	—	0	—	nS
T _{DS}	Data Bus Setup Time	150	—	75	—	nS
T _{DH}	Data Bus Hold Time	0	—	0	—	nS
Read Cycle						
T _{PS}	Port Setup Time	300	—	150	—	nS
T _{WCS(2)}	CS low while ϕ_2 high	420	—	20	—	nS
T _{ADS}	Address Setup Time	0	—	0	—	nS
T _{ADH}	Address Hold Time	10	—	5	—	nS
T _{RWS}	R/W Setup Time	0	—	0	—	nS
T _{RWH}	R/W Hold Time	0	—	0	—	nS
T _{ACC}	Data Access from RS3-RS0	—	550	—	275	nS
T _{CO(3)}	Data Access from CS	—	320	—	150	nS
T _{DR}	Data Release Time	50	—	25	—	nS

NOTES: 1 — All timings are referenced from V_{IL} max and V_{IH} min on inputs and V_{OL} max and V_{OH} min on outputs.
2 — T_{WCS} is measured from the later of ϕ_2 high or CS low. CS must be low at least until the end of ϕ_2 high.
3 — T_{CO} is measured from the later of ϕ_2 high or CS low.
Valid data is available only after the later of T_{ACC} or T_{CO}.

REGISTER MAP

RS3	RS2	RS1	RS0	REG	
0	0	0	0	0	PRA PERIPHERAL DATA REG A
0	0	0	1	1	PRB PERIPHERAL DATA REG B
0	0	1	0	2	DDRA DATA DIRECTION REG A
0	0	1	1	3	DDRB DATA DIRECTION REG B
0	1	0	0	4	TA LO TIMER A LOW REGISTER
0	1	0	1	5	TA HI TIMER A HIGH REGISTER
0	1	1	0	6	TB LO TIMER B LOW REGISTER
0	1	1	1	7	TB HI TIMER B HIGH REGISTER
1	0	0	0	8	TOD 10THS 10THS OF SECONDS REGISTER
1	0	0	1	9	TOD SEC SECONDS REGISTER
1	0	1	0	A	TOD MIN MINUTES REGISTER
1	0	1	1	B	TOD HR HOURS — AM/PM REGISTER
1	1	0	0	C	SDR SERIAL DATA REGISTER
1	1	0	1	D	ICR INTERRUPT CONTROL REGISTER
1	1	1	0	E	CRA CONTROL REG A
1	1	1	1	F	CRB CONTROL REG B

6526 FUNCTIONAL DESCRIPTION

I/O Ports (PRA, PRB, DDRA, DDRB).

Ports A and B each consist of an 8-bit Peripheral Data Register (PR) and an 8-bit Data Direction Register (DDR). If a bit in the DDR is set to the corresponding bit in the PR is an output, if a DDR bit is set to a zero, the corresponding PR bit is defined as an input. On a READ, the PR reflects the information present on the actual port pins (PA0-PA7, PB0-PB7) for both input and output bits. Port A and Port B have passive pull-up devices as well as active pull-ups, providing both CMOS and TTL compatibility. Both ports have two TTL load drive capability. In addition to normal I/O operation, PB6 and PB7 also provide timer output functions.

Handshaking

Handshaking on data transfers can be accomplished using the $\overline{PC}$ output pin and the $\overline{FLAG}$ input pin. $\overline{PC}$ will go low for one cycle following a read or write of PORT B. This signal can be used to indicate "data ready" at PORT B or "data accepted" from PORT B. Handshaking on 16-bit data transfers (using both PORT A and PORT B) is possible by always reading or writing PORT A first. $\overline{FLAG}$ is a negative edge sensitive input which can be used for receiving the $\overline{PC}$ output from another 6526, or as a general purpose interrupt input. Any negative transition on $\overline{FLAG}$ will set the $\overline{FLAG}$ interrupt bit.

REG	NAME	D ₇	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀
0	PRA	PA ₇	PA ₆	PA ₅	PA ₄	PA ₃	PA ₂	PA ₁	PA ₀
1	PRB	PB ₇	PB ₆	PB ₅	PB ₄	PB ₃	PB ₂	PB ₁	PB ₀
2	DDRA	DPA ₇	DPA ₆	DPA ₅	DPA ₄	DPA ₃	DPA ₂	DPA ₁	DPA ₀
3	DDRB	DPB ₇	DPB ₆	DPB ₅	DPB ₄	DPB ₃	DPB ₂	DPB ₁	DPB ₀

Interval Timers (Timer A, Timer B)

Each interval timer consists of a 16-bit read-only Timer Counter and a 16-bit write-only Timer Latch. Data written to the timer are latched in the Timer Latch, while data read from the timer are the present contents of the Timer Counter. The timers can be used independently or linked for extended operations. The various timer modes allow generation of long time delays, variable width pulses, pulse trains and variable frequency waveforms. Utilizing the CNT input, the timers can count external pulses or measure frequency, pulse width and delay times of external signals. Each timer has an associated control register, providing independent control of the following functions:

Start/Stop

A control bit allows the timer to be started or stopped by the microprocessor at any time.

PB On/Off:

A control bit allows the timer output to appear on a PORT B output line (PB6 for TIMER A and PB7 for TIMER B). This function overrides the DDRB control bit and forces the appropriate PB line to an output.

Toggle/Pulse

A control bit selects the output applied to PORT B. On every timer underflow the output can either toggle or generate a single positive pulse of one cycle duration. The Toggle output is set high whenever the timer is started and is set low by $\overline{RES}$.

One-Short/Continuous

A control bit selects either timer mode. In one-shot mode, the timer will count down from the latched value to zero, generate an interrupt, reload the latched value, then stop. In continuous mode, the timer will count from the latched value to zero, generate an interrupt, reload the latched value and repeat the procedure continuously.

Force Load

A strobe bit allows the timer latch to be loaded into the timer counter at any time, whether the timer is running or not.

Input Mode:

Control bits allow selection of the clock used to decrement the timer. TIMER A can count $\emptyset 2$ clock pulses or external pulses applied to the CNT pin. TIMER B can count $\emptyset 2$ pulses, external CNT pulses, TIMER A underflow pulses or TIMER A underflow pulses while the CNT pin is held high.

The timer latch is loaded into the timer on any timer underflow, on a force load or following a write to the high byte of the prescaler while the timer is stopped. If the timer is running, a write to the high byte will load the timer latch, but not reload the counter.

READ (TIMER)

REG	NAME	TAL ₇	TAL ₆	TAL ₅	TAL ₄	TAL ₃	TAL ₂	TAL ₁	TAL ₀
4	TA LO	TAL ₇	TAL ₆	TAL ₅	TAL ₄	TAL ₃	TAL ₂	TAL ₁	TAL ₀
5	TA HI	TAH ₇	TAH ₆	TAH ₅	TAH ₄	TAH ₃	TAH ₂	TAH ₁	TAH ₀
6	TB LO	TBL ₇	TBL ₆	TBL ₅	TBL ₄	TBL ₃	TBL ₂	TBL ₁	TBL ₀
7	TB HI	TBH ₇	TBH ₆	TBH ₅	TBH ₄	TBH ₃	TBH ₂	TBH ₁	TBH ₀

WRITE (PRESCALER)

REG	NAME	PAL ₇	PAL ₆	PAL ₅	PAL ₄	PAL ₃	PAL ₂	PAL ₁	PAL ₀
4	TA LO	PAL ₇	PAL ₆	PAL ₅	PAL ₄	PAL ₃	PAL ₂	PAL ₁	PAL ₀
5	TA HI	PAH ₇	PAH ₆	PAH ₅	PAH ₄	PAH ₃	PAH ₂	PAH ₁	PAH ₀
6	TB LO	PBL ₇	PBL ₆	PBL ₅	PBL ₄	PBL ₃	PBL ₂	PBL ₁	PBL ₀
7	TB HI	PBH ₇	PBH ₆	PBH ₅	PBH ₄	PBH ₃	PBH ₂	PBH ₁	PBH ₀

Time of Day Clock (TOD)

The TOD clock is a special purpose timer for real-time applications. TOD consists of a 24-hour (AM/PM) clock with 1/10th second resolution. It is organized into 4 registers: 10ths of seconds, Seconds, Minutes and Hours. The AM/PM flag is in the MSB of the Hours register for easy bit testing. Each register reads out in BCD format to simplify conversion for driving displays, etc. The clock requires an external 60 Hz or 50 Hz (programmable) TTL level input on the TOD pin for accurate time-keeping. In addition to time-keeping, a programmable ALARM is provided for generating an interrupt at a desired time. The ALARM registers are located at the same addresses as the corresponding TOD registers. Access to the ALARM is governed by a Control Register bit. The ALARM is write-only; any read of a TOD address will read time regardless of the state of the ALARM access bit.

A specific sequence of events must be followed for proper setting and reading of TOD. TOD is automatically stopped whenever a write to the Hours register occurs. The clock will not start again until after a write to the 10ths of seconds register. This assures TOD will always start at the desired time. Since a carry

from one stage to the next can occur at any time with respect to a read operation, a latching function is included to keep all Time Of Day information constant during a read sequence. All four TOD registers latch on a read of Hours and remain latched until after a read of 10ths of seconds. The TOD clock continues to count when the output registers are latched. If only one register is to be read, there is no carry problem and the register can be read "on the fly," provided that any read of Hours is followed by a read of 10ths of seconds to disable the latching.

READ

REG NAME

8	TOD 10THS	0	0	0	0	T ₈	T ₄	T ₂	T ₁
9	TOD SEC	0	SH ₄	SH ₂	SH ₁	SL ₈	SL ₄	SL ₂	SL ₁
A	TOD MIN	0	MH ₄	MH ₂	MH ₁	ML ₈	ML ₄	ML ₂	ML ₁
B	TOD HR	PM	0	0	HH	HL ₈	HL ₄	HL ₂	HL ₁

WRITE

CRB₇=0 TOD

CRB₇=1 ALARM

(SAME FORMAT AS READ)

Serial Port (SDR)

The serial port is a buffered, 8-bit synchronous shift register system. A control bit selects input or output mode. In input mode, data on the SP pin is shifted into the shift register on the rising edge of the signal applied to the CNT pin. After 8 CNT pulses, the data in the shift register is dumped into the Serial Data Register and an interrupt is generated. In the output mode, TIMER A is used for the baud rate generator. Data is shifted out on the SP pin at 1/2 the underflow rate of TIMER A. The maximum baud rate possible is 2 divided by 4, but the maximum useable baud rate will be determined by line loading and the speed at which the receiver responds to input data. Transmission will start following a write to the Serial Data Register (provided TIMER A is running and in continuous mode). The clock signal derived from TIMER A appears as an output on the CNT pin. The data in the Serial Data Register will be loaded into the shift register then shift out to the SP pin when a CNT pulse occurs. Data shifted out becomes valid on the falling edge of CNT and remains valid until the next falling edge. After 8 CNT pulses, an interrupt is generated to indicate more data can be sent. If the Serial Data Register was loaded with new information prior to this interrupt, the new data will automatically be loaded into the shift register and transmission will continue. If the microprocessor stays one byte ahead of the shift register, transmission will be continuous. If no further data is to be transmitted, after the 8th CNT pulse, CNT will return high and SP will remain at the level of the last data bit transmitted. SDR data is shifted out MSB first and serial input data should also appear in this format.

The bidirectional capability of the Serial Port and CNT clock allows many 6526 devices to be connected to a common serial communication bus on which one 6526 acts as a master, sourcing data and shift clock, while all other 6526 chips act as slaves. Both CNT and SP outputs are open drain to allow such a common bus. Protocol for master/slave selection can be transmitted over the serial bus, or via dedicated handshaking lines.

REG NAME

C	SDR	S ₇	S ₆	S ₅	S ₄	S ₃	S ₂	S ₁	S ₀
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Interrupt Control (ICR)

There are five sources of interrupts on the 6526: underflow from TIMER A, underflow from TIMER B, TOD ALARM, Serial Port full/empty and FLAG. A single register provides masking and interrupt information. The Interrupt Control Register consists of a write-only MASK register and a read-only DATA register. Any interrupt will set the corresponding bit in the DATA register. Any interrupt which is enabled by the MASK register will set the IR bit (MSB) of the DATA register and bring the $\overline{\text{IRQ}}$ pin low. In a multi-chip system, the IR bit can be polled to detect which chip has generated an interrupt request. The interrupt DATA register is cleared and the $\overline{\text{IRQ}}$ line returns high following a read of the DATA register. Since each interrupt sets an interrupt bit regardless of the MASK, and each interrupt bit can be selectively masked to prevent the generation of a processor interrupt, it is possible to intermix polled interrupts with true interrupts. However, polling the IR bit will cause the DATA register to clear, therefore, it is up to the user to preserve the information contained in the DATA register if any polled interrupts were present.

The MASK register provides convenient control of individual mask bits. When writing to the MASK register, if bit 7 (SET/CLEAR) of the data written is a ZERO, any mask bit written with a one will be cleared, while those mask bits written with a zero will be unaffected. If bit 7 of the data written is a ONE, any mask bit written with a one will be set, while those mask bits written with a zero will be unaffected. In order for an interrupt flag to set IR and generate an Interrupt Request, the corresponding MASK bit must be set.

READ (INT DATA)

REG NAME

D	ICR	IR	0	0	FLG	SP	ALRM	TB	TA
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WRITE (INT MASK)

REG NAME

D	ICR	S/C	X	X	FLG	SP	ALRM	TB	TA
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CONTROL REGISTERS

There are two control registers in the 6526, CRA and CRB. CRA is associated with TIMER A and CRB is associated with TIMER B. The register format is as follows:

CRA:

Bit	Name	Function
0	START	1=START TIMER A, 0=STOP TIMER A. This bit is automatically reset when underflow occurs during one-shot mode.
1	PBON	1=TIMER A output appears on PB6, 0=PB6 normal operation.
2	OUTMODE	1=TOGGLE, 0=PULSE
3	RUNMODE	1=ONE-SHOT, 0=CONTINUOUS
4	LOAD	1=FORCE LOAD (this is a STROBE input, there is no data storage, bit 4 will always read back a zero and writing a zero has no effect).
5	INMODE	1=TIMER A counts positive CNT transitions, 0=TIMER A counts $\emptyset$ 2 pulses.
6	SPMODE	1=SERIAL PORT output (CNT sources shift clock), 0=SERIAL PORT input (external shift clock required).
7	TODIN	1=50 Hz clock required on TOD pin for accurate time, 0=60 Hz clock required on TOD pin for accurate time.

CRB:

Bit Name	Function															
	(Bits CRB0-CRB4 are identical to CRA0-CRA4 for TIMER B with the exception that bit 1 controls the output of TIMER B on PB7).															
5,6 INMODE	Bits CRB5 and CRB6 select one of four input modes for TIMER B as:															
	<table><tr><th>CRB6</th><th>CRB5</th><th></th></tr><tr><td>0</td><td>0</td><td>TIMER B counts $\varnothing$2 pulses.</td></tr><tr><td>0</td><td>1</td><td>TIMER B counts positive CNT transistions.</td></tr><tr><td>1</td><td>0</td><td>TIMER B counts <i>TIMER A underflow</i> pulses.</td></tr><tr><td>1</td><td>1</td><td>TIMER B counts <i>TIMER A underflow</i> pulses while CNT is high.</td></tr></table>	CRB6	CRB5		0	0	TIMER B counts $\varnothing$ 2 pulses.	0	1	TIMER B counts positive CNT transistions.	1	0	TIMER B counts <i>TIMER A underflow</i> pulses.	1	1	TIMER B counts <i>TIMER A underflow</i> pulses while CNT is high.
CRB6	CRB5															
0	0	TIMER B counts $\varnothing$ 2 pulses.														
0	1	TIMER B counts positive CNT transistions.														
1	0	TIMER B counts <i>TIMER A underflow</i> pulses.														
1	1	TIMER B counts <i>TIMER A underflow</i> pulses while CNT is high.														
7 ALARM	1=writing to TOD registers sets ALARM, 0=writing to TOD registers sets TOD clock.															

REG	NAME	TOD IN	SP MODE	IN MODE	LOAD	RUN MODE	OUT MODE	PB ON	START
E	CRA	0=60Hz 1=50Hz	0=INPUT 1=OUTPUT	0= ϕ_2 1=CNT	1=FORCE LOAD (STROBE)	0=CONT. 1=O.S.	0=PULSE 1=TOGGLE	0=PB ₆ OFF 1=PB ₆ ON	0=STOP 1=START

REG	NAME	ALARM	IN MODE	LOAD	RUN MODE	OUT MODE	PB ON	START
F	CRB	0=TOD 1=ALARM	0 0 1 1	0=02 1=CNT 0=TA 1=CNT•TA	1=FORCE LOAD (STROBE)	0=CONT. 1=O.S.	0=PULSE 1=TOGGLE	0=PB7 OFF 1=PB7 ON

All unused register bits are unaffected by a write and are forced to zero on a read.

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